COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010/Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	July 9, 2026	**DATE SUBMITTED:**	July 14, 2026
NAME:	Jaime D.C. Godino Jr.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	
Class Declaration and Method Definition Section	
Main Function	

```cpp
//Create a function that will take two numbers and display the sum

int sumOfTwoNum(int fNum, int sNum);
void sumOfThreeNum(int fNum, int sNum, int tNum);


//Create a function that will return whether variable A is greater than variable

bool greaterNum(int A, int B);

//Create a function that will take two Boolean values and display the result of all
//logical operations return true if it was a success

bool logicalOperators(bool a, bool b);
```

```cpp
int main()
{
    std::cout<<"The sum of the two number is: "<<sumOfTwoNum(6,7)<<std::endl;
    sumOfThreeNum(6,7,8);

    int A, B;

    std::cout <<"Comparing greater value:"<< std::endl;
    greaterNum(7, 8);

    if (greaterNum(A,B)){
        std::cout <<"A is greater than B"<< std::endl;
    }else{
        std::cout <<"A is NOT greater than B"<< std::endl;
    }

    std::cout<< "The result of the two boolean value is: \n" << logicalOperators(1,0);


}
```

Method Definition	

```cpp
int sumOfTwoNum(int fNum, int sNum){

    int sum = fNum + sNum;

    return sum;
}
void sumOfThreeNum(int fNum, int sNum, int tNum){

    std::cout<< "sum: " << fNum+sNum+tNum <<std::endl;
}
bool greaterNum(int A, int B){
    return A > B;
}
bool logicalOperators(bool a, bool b){
    std::cout<<"AND: " << (a && b)<< std::endl;
    std::cout<<"OR : " << (a || b)<< std::endl;
    std::cout<<"NOT: " << (!a)<< std::endl;
    std::cout<<"NOT: " << (!b)<< std::endl;

    return true;
}
```

Table 1 C++ Structure Code for Answer

```
The sum of the two number is: 13
sum: 21
Comparing greater value:
A is NOT greater than B
The result of the two boolean value is:
AND: 0
OR : 1
NOT: 0
NOT: 1
1
```

Output Observations and Comments:

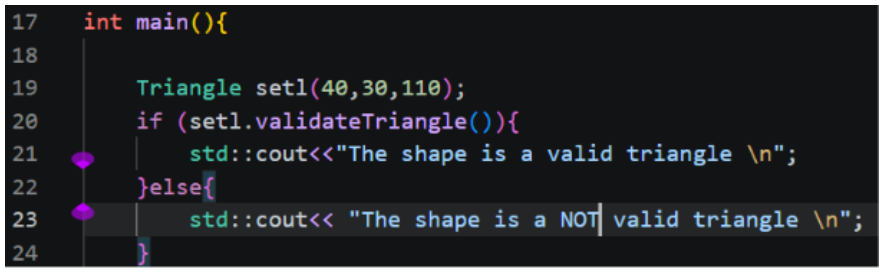
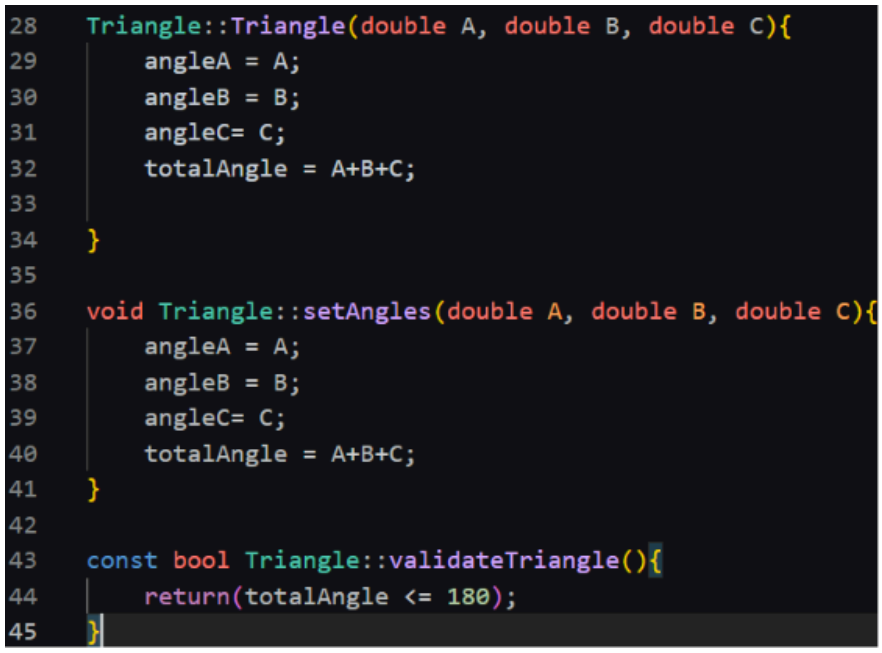
The program correctly computes and displays the sum of two numbers and the sum of three numbers, producing the values 13 and 21, respectively. It also successfully compares two values and correctly identifies that A is greater than B. Lastly, the program performs the basic Boolean logical operations and displays the expected results, showing that the logical function works as intended.

Table 1: ILO A output observations and comments.

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	N/A

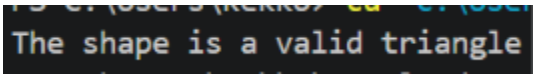
Class Declaration and Method Definition Section	`class Triangle{`
	`private:`
	`double totalAngle, angleA, angleB, angleC;`
	`public:`
	`//constructor`
	`Triangle(double A, double B, double C);`
	`void setAngles(double A, double B, double C);`
	
	`//triangle validation`
	`const bool validateTriangle();`

Main Function	
Method Definition	

```
17  int main(){
18
19      Triangle set1(40,30,110);
20      if (set1.validateTriangle()){
21          std::cout<<"The shape is a valid triangle \n";
22      }else{
23          std::cout<< "The shape is a NOT valid triangle \n";
24      }
```

```
28  Triangle::Triangle(double A, double B, double C){
29      angleA = A;
30      angleB = B;
31      angleC= C;
32      totalAngle = A+B+C;
33
34  }
35
36  void Triangle::setAngles(double A, double B, double C){
37      angleA = A;
38      angleB = B;
39      angleC= C;
40      totalAngle = A+B+C;
41  }
42
43  const bool Triangle::validateTriangle(){
44      return(totalAngle <= 180);
45  }
```

Table 2 C++ Structure Code for Answer



Output Observations and Comments:
The program correctly outputs "The shape is a valid triangle" because the three angles add up to 180°, satisfying the validation condition. This demonstrates that the constructor initializes the angle values correctly and that the validateTriangle() method successfully checks the total angle.

Table 2: ILO B output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve Different Problems using the C++ Programming Language

Code:

```cpp
#include <iostream>
#include <cmath>
#include <utility>

class SwapNumbers {
private:
    int num1, num2;

public:
    void input();
    void swapNumbers();
};

class ConvertTemperature {
private:
    double kelvin;

public:
    void inputTemperature();
    double convertTemperature();
};

class CalculateDistance {
private:
    double x1, y1, x2, y2;

public:
    CalculateDistance(double, double, double, double);
    double calculateDistance();
};

class Triangle {
private:
    double angleA, angleB, angleC;
    double sideA, sideB, sideC;

public:
    Triangle(double, double, double, double, double, double);

    bool validateTriangle();
    double computeArea();
    double computePerimeter();
    void triangleType();
};
```

```
int main()
{
    SwapNumbers swap;
    swap.input();
    swap.swapNumbers();

    ConvertTemperature converter;
    converter.inputTemperature();
    std::cout << "\nTemperature in Fahrenheit: "
              << converter.convertTemperature() << " F\n";

    CalculateDistance points(2, 3, 6, 7);
    std::cout << "\nDistance between the two points: "
              << points.calculateDistance() << std::endl;

    Triangle t(40, 30, 110, 5, 6, 7);

    if (t.validateTriangle()) {
        std::cout << "\nThe shape is a valid triangle.\n";
        std::cout << "Area = " << t.computeArea() << std::endl;
        std::cout << "Perimeter = " << t.computePerimeter() << std::endl;
        t.triangleType();
    }
    else {
        std::cout << "\nThe shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

```
void SwapNumbers::input()
{
    std::cout << "Enter first number: ";
    std::cin >> num1;
    std::cout << "Enter second number: ";
    std::cin >> num2;

    std::cout << "\nBefore Swapping\n";
    std::cout << "First Number = " << num1 << "\n";
    std::cout << "Second Number = " << num2 << "\n";
}

void SwapNumbers::swapNumbers()
{
    std::swap(num1, num2);

    std::cout << "\nAfter Swapping\n";
    std::cout << "First Number = " << num1 << "\n";
    std::cout << "Second Number = " << num2 << "\n";
}

void ConvertTemperature::inputTemperature()
```

```
98  {
99      std::cout << "\nEnter temperature in Kelvin: ";
100     std::cin >> kelvin;
101 }
102
103 double ConvertTemperature::convertTemperature()
104 {
105     return (kelvin - 273.15) * 9 / 5 + 32;
106 }
107
108 CalculateDistance::CalculateDistance(double a, double b, double c, double d)
109     : x1(a), y1(b), x2(c), y2(d) {}
110
111 double CalculateDistance::calculateDistance()
112 {
113     return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
114 }
115
116 Triangle::Triangle(double A, double B, double C,
117                    double a, double b, double c)
118     : angleA(A), angleB(B), angleC(C),
119       sideA(a), sideB(b), sideC(c) {}
120
121 bool Triangle::validateTriangle()
```

```
122 {
123     return (angleA + angleB + angleC == 180);
124 }
125
126 double Triangle::computePerimeter()
127 {
128     return sideA + sideB + sideC;
129 }
130
131 double Triangle::computeArea()
132 {
133     double s = computePerimeter() / 2;
134     return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
135 }
136
137 void Triangle::triangleType()
138 {
139     if (angleA < 90 && angleB < 90 && angleC < 90)
140         std::cout << "Triangle Type: Acute-angled\n";
141     else if (angleA > 90 || angleB > 90 || angleC > 90)
142         std::cout << "Triangle Type: Obtuse-angled\n";
143     else
144         std::cout << "Triangle Type: Others\n";
145 }
```

Output

```
Enter first number: 10
Enter second number: 15

Before Swapping
First Number = 10
Second Number = 15

After Swapping
First Number = 15
Second Number = 10

Enter temperature in Kelvin: 20

Temperature in Fahrenheit: -423.67 F

Distance between the two points: 5.65685

The shape is a valid triangle.
Area = 14.6969
Perimeter = 18
Triangle Type: Obtuse-angled
```

Conclusion:

The supplementary activity allowed me to use and develop classes that perform swapping numbers, converting temperature, calculating the distance between two points, and validating and analyzing a triangle. I also used constructors, methods, encapsulation, and computation to solve the given problems. Overall, I was able to finish the activity with a deeper understanding, although I still need more practice optimizing my code and handling more complex programming scenarios with time efficiency since the whole activity took me a while to finish.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT, Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I asked ChatGPT how to shorten my code since it was formerly around 200+ lines of code. It did not replace my original thinking since I wrote the code myself and only asked ChatGPT to shorten it so it would look clean. Finally, I wrote all my answers while Grammarly corrected my grammar mistakes**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the

permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.